
ASSIGMENT

PROGRAMMING USING C

WEEK – 03

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Week – 3(a)

Write a program to read two integer values and print true if both the numbers end with the same digit, otherwise print false.

Example: If 698 and 768 are given, program should print true as they both end with 8. Sample Input 1 25 53 Sample Output 1 false

Sample Input 2 27 77 Sample Output 2 true

```
#include<stdio.h>
int main()
{
    int a,b,c,d;
    scanf("%d%d",&a,&b);
    c=a%10;
    d=b%10;
    if(c==d)
        printf("true");
    else
        printf("false");
}
```

	Input	Expected	Got	
✓	25 53	false	false	✓
✓	27 77	true	true	✓

Passed all tests! ✓

Given an integer, n , perform the following conditional actions:

- If n is odd, print **Weird**
- If n is even and in the inclusive range of **2** to **5**, print **Not Weird**
- If n is even and in the inclusive range of **6** to **20**, print **Weird**
- If n is even and greater than **20**, print **Not Weird**

Complete the stub code provided in your editor to print whether or not n is weird.

```
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    if(n%2==0)
    {
        if(n>=2&& n<=5)
        {
            printf("Not Weird");
        }
        else if(n>=6&& n<=20)
        {
            printf("Weird");
        }
        else
        {
            printf("Not Weird");
        }
    }
    else
    {
        printf("Weird");
    }
}
```

	Input	Expected	Got	
✓	3	Weird	Weird	✓
✓	24	Not Weird	Not Weird	✓

Passed all tests! ✓

Three numbers form a Pythagorean triple if the sum of squares of two numbers is equal to the square of the third. For example, 3, 5 and 4 form a Pythagorean triple, since $3^2 + 4^2 = 25 = 5^2$. You are given three integers, a, b, and c. They need not be given in increasing order. If they form a Pythagorean triple, then print "yes", otherwise, print "no". Please note that the output message is in small letters. Sample Input 1 3 5 4 Sample Output 1 yes Sample Input 2 5 8 2 Sample Output 2 no

```
#include<stdio.h>
int main()
{
    int a,b,c;
    scanf("%d%d%d",&a,&b,&c);
    a=a*a;
    b=b*b;
    c=c*c;
    if((a+b==c) || (a+c==b) || (b+c==a))
    {
        printf("yes");
    }
    else
    {
        printf("no");
    }
}
```

	Input	Expected	Got	
✓	3 5 4	yes	yes	✓
✓	5 8 2	no	no	✓

Passed all tests! ✓

Week – 3(b)

Write a program that determines the name of a shape from its number of sides. Read the number of sides from the user and then report the appropriate name as part of a meaningful message. Your program should support shapes with anywhere from 3 up to (and including) 10 sides. If a number of sides outside of this range is entered then your program should display an appropriate error message.

```
#include<stdio.h>
int main()
{
    int a;
    scanf("%d",&a);
    switch(a)
    {
        case 3:
            printf("Triangle");
            break;
        case 4:
            printf("Square");
            break;
        case 5:
            printf("Pentagon");
            break;
        case 6:
            printf("Hexagon");
            break;
        case 7:
            printf("Heptagon");
            break;
        case 8:
            printf("Octagon");
            break;
        case 9:
            printf("Nonagon");
            break;
        default:
            {
                printf("The number of sides is not supported.");
            }
    }
}
```

	Input	Expected	Got	
✓	3	Triangle	Triangle	✓
✓	7	Heptagon	Heptagon	✓
✓	11	The number of sides is not supported.	The number of sides is not supported.	✓

Passed all tests! ✓

The Chinese zodiac assigns animals to years in a 12-year cycle. One 12-year cycle is shown in the table below. The pattern repeats from there, with 2012 being another year of the Dragon, and 1999 being another year of the Hare.

Year	Animal
2000	Dragon
2001	Snake
2002	Horse
2003	Sheep
2004	Monkey
2005	Rooster
2006	Dog
2007	Pig
2008	Rat
2009	Ox
2010	Tiger
2011	Hare

```
#include<stdio.h>
int main()
{
    int a;
    scanf("%d",&a);
    if(a==2000||(2000-a)%12==0)
        printf("Dragon");
    if(a==2001||(2001-a)%12==0)
        printf("Snake");
    if(a==2002||(2002-a)%12==0)
        printf("Horse");
    if(a==2003||(2003-a)%12==0)
        printf("Sheep");
    if(a==2004||(2004-a)%12==0)
        printf("Monkey");
    if(a==2005||(2005-a)%12==0)
        printf("Rooster");
    if(a==2006||(2006-a)%12==0)
        printf("Dog");
    if(a==2007||(2007-a)%12==0)
        printf("Pig");
    if(a==2008||(2008-a)%12==0)
        printf("Rat");
    if(a==2009||(2009-a)%12==0)
        printf("Ox");
    if(a==2010||(2010-a)%12==0)
        printf("Tiger");
    if(a==2011||(2011-a)%12==0)
        printf("Hare");
    return 0;
}
```

	Input	Expected	Got	
✓	2004	Monkey	Monkey	✓
✓	2010	Tiger	Tiger	✓

Passed all tests! ✓

Write a program that reads a position from the user. Use an if statement to determine if the column begins with a black square or a white square. Then use modular arithmetic to report the color of the square in that row. For example, if the user enters a1 then your program should report that the square is black. If the user enters d5 then your program should report that the square is white. Your program may assume that a valid position will always be entered. It does not need to perform any error checking.

```
#include<stdio.h>
int main()
{
    int n;
    char a;
    scanf("%c%d",&a,&n);
    if(a=='a' || a=='c' || a=='e' || a=='g')
    {
        if (n%2!=0)
        {
            printf("The square is black.");
        }
        else
        {
            printf("The square is white.");
        }
    }
    else if(a=='b' || a=='d' || a=='f' || a=='h' )
    {
        if (n%2!=0)
        {
            printf("The square is white.");
        }
        else
        {
            printf("The square is black.");
        }
    }
}
```

	Input	Expected	Got	
✓	a 1	The square is black.	The square is black.	✓
✓	d 5	The square is white.	The square is white.	✓

Passed all tests! ✓

Week – 3(c)

Some data sets specify dates using the year and day of year rather than the year, month, and day of month. The day of year (DOY) is the sequential day number starting with day 1 on January 1st.

```
#include<stdio.h>
int main()
{
    int d,m,y,feb;
    scanf("%d%d%d",&d,&m,&y);
    if((y%100==0&& y%400) || (y%4==0))
        feb=29;
    else
        feb=28;
    switch(m)
    {
        case 1:
            printf("%d",d);
            break;
        case 2:
            printf("%d",31+d);
            break;
        case 3:
            printf("%d",31+feb+d);
            break;
        case 4:
            printf("%d",31+feb+31+d);
            break;
        case 5:
            printf("%d",31+feb+31+30+d);
            break;
        case 6:
            printf("%d",31+feb+31+30+31+d);
            break;
        case 7:
            printf("%d",31+feb+31+30+31+30+d);
            break;
        case 8:
            printf("%d",31+feb+31+30+31+30+31+d);
            break;
        case 9:
            printf("%d",31+feb+31+30+31+30+31+31+d);
            break;
        case 10:
            printf("%d",31+feb+31+30+31+30+31+31+30+d);
            break;
        case 11:
            printf("%d",31+feb+31+30+31+30+31+31+30+31+d);
            break;
        case 12:
            printf("%d",31+feb+31+30+31+30+31+31+30+31+30+d);
            break;
    }
}
```

	Input	Expected	Got	
✓	18 6 2020	170	170	✓

Suppandi is trying to take part in the local village math quiz. In the first round, he is asked about shapes and areas. Suppandi, is confused, he was never any good at math. And also, he is bad at remembering the names of shapes. Instead, you will be helping him [calculate the area](#) of shapes.

```
#include<stdio.h>
int main()
{
    int a,b,l;
    char c;
    scanf("%c%d%d",&c,&b,&l);
    if(c=='R')
    {
        a=l*b;
        printf("%d",a);
    }
    else if(c=='S')
    {
        a=(0.5)*l*b;
        printf("%d",a);
    }
    else if(c=='T')
    {
        a=l*b;
        printf("%d",a);
    }
    else
        printf("0");
}
```

	Input	Expected	Got	
✓	T 10 20	200	200	✓
✓	S 30 40	600	600	✓
✓	B 2 11	0	0	✓
✓	R 10 30	300	300	✓
✓	S 40 50	1000	1000	✓
Passed all tests! ✓				

Superman is planning a journey to his home planet. It is very important for him to know which day he arrives there. They don't follow the 7-day week like us. Instead, they follow a 10-day week with the following days: Day Number Name of Day 1 Sunday 2 Monday 3 Tuesday 4 Wednesday 5 Thursday 6 Friday 7 Saturday 8 Kryptonday 9 Coluday 10 Daxamday Here are the rules of the calendar:

- The calendar starts with Sunday always.
- It has only 296 days. After the 296th day, it goes back to Sunday. You begin your journey on a Sunday and will reach after n. You have to tell on which day you will arrive when you reach there.

```
#include<stdio.h>
int main()
{
    int n,day;
    scanf("%d",&n);
    if(n<296)
        day=n;
    else
        day=n-296;
    day=day%10;
    day+=1;
    switch(day)
    {
        case 1:
            printf("Sunday");
            break;
        case 2:
            printf("Monday");
            break;
        case 3:
            printf("Tuesday");
            break;
        case 4:
            printf("Wednesday");
            break;
        case 5:
            printf("Thursday");
            break;
        case 6:
            printf("Friday");
            break;
        case 7:
            printf("Saturday");
            break;
        case 8:
            printf("Kryptonday");
            break;
        case 9:
            printf("coluday");
            break;
        case 10:
            printf("Daxamday");
            break;
    }
    return 0;
}
```

	Input	Expected	Got	
✓	7	Kryptonday	Kryptonday	✓
✓	1	Monday	Monday	✓

Passed all tests! ✓